Lecture 1 · Robots That Do the Task

Integrated Intelligent Systems · student chapter · companion to the revised 32-slide lecture

An intelligent robot is judged by what it can accomplish in the world. A request such as “bring me my coffee” sounds simple because a person supplies much of the missing information without noticing. The robot must identify the intended outcome, interpret the present situation, choose actions its body can perform, detect when its assumptions fail, and establish whether the outcome was achieved. This chapter develops the question that will organize the course: **which control programs can solve which classes of manipulation problems, with which bodies, in which worlds, and on what evidence?**

The chapter follows the five parts of the lecture. Slide numbers refer to the current 32-slide deck. The examples of Eva, the production cell, and the laboratory are **illustrative task scenarios** unless a demonstration is explicitly identified. Numerical details in these scenarios are teaching assumptions, not measurements from deployed robots.

---

1. From a request to a physical task · slides 1–10

Eva asks for coffee

Imagine that Eva says to a household robot, “Bring me my coffee.” The desired end is that Eva receives a suitable drink, within reach, without being harmed or obstructed. The sentence does not specify the drink's identity, which mug to use, where the mug is now, how full it is, which route is clear, or how Eva wants the handover. These omissions matter because the robot's movements will change a real room containing a person and fragile objects.

Suppose Eva normally uses a blue mug, but it is in the dishwasher. A white mug is available. The robot cannot infer from availability alone that the substitute is acceptable. It can ask Eva. Suppose she accepts the white mug. The robot still needs to determine its fill, a safe way to carry it, and a handover position that remains appropriate if Eva stands or moves her walking frame. One short request has become a sequence of *grounding*, *prediction*, *action*, and *checking* decisions. Grounding means connecting words such as “my coffee” to the particular objects, person, constraints, and desired outcome in this situation.

The goal can remain stable while the method changes. If the blue mug is unavailable, asking about a substitute is part of pursuing the goal. If the white mug is nearly full, the robot may need to slow a turn or use a tray. If Eva stands, it may need to pause and choose a new handover. Flexibility is therefore more than substituting one motion for another. The system must work out what success means in this context and preserve the goal as the context changes.

> **Rehearsal pause.** For Eva's request, write down one desired outcome, two constraints, and two facts the robot still needs to establish. Then state which of your facts could change during execution.

Why movement and things matter

The lecture uses the sea squirt and the competence ladder as illustrations of a course perspective: a brain helps a body move so that the world changes in pursuit of a goal. This is a useful way to study embodied intelligence, not an exhaustive biological account of brains. The ladder moves from surviving and moving to adapting, using things, understanding, and planning. The ape using a tool is the crucial image. A tool changes the animal's *effective embodiment*: it extends what can be reached or manipulated. Using the tool is a means; obtaining a desired change in the world is the end.

The same distinction matters for robots. A gripper, tray, mobile base, or external fixture changes what actions are feasible. The robot must choose and use these resources in relation to its goal, not merely execute movements that look skilled. The grand challenge is to accomplish **changing, partly specified tasks in an open world**, where the meaning of success depends on the situation. Novel goals and novel contexts demand reliable transfer, monitoring, and revision.

A robot has a body with links, joints, actuators, sensors, and often a mobile base. A control program closes a loop: it uses observations to choose an action, the body acts on the world, and new observations inform the next choice. A precomputed path alone cannot handle a displaced mug or a moving recipient. The *body motion problem*, developed in Lecture 2, asks how a task request becomes physical motions that produce the wanted effect while avoiding unwanted effects.

The challenge and three settings

Nils Nilsson's employment test asks us to consider jobs a machine can perform acceptably. The lecture adapts that challenge to a robot: imagine a skilled operator using the robot's own cameras, arms, gripper, and mobility through a game controller. What tasks could that person complete? The table-setting video on slide 7 illustrates a physical household task. Operator success shows something about the body and the operator's judgment under those conditions. It does not establish that an autonomous program has the sensing, interpretation, and control needed to do the task by itself, or that it can repeat it across varied homes.

The challenge appears in three application settings. In an **illustrative production task**, a robot inserts a measured stator segment into its matching slot. It must align the part, regulate contact, protect the insulation, and verify the result. Slide 9 assumes a force limit of 340 newtons and a segment measured at plus 0.08 millimeters. These figures are teaching assumptions, not measured robot data. A small difference between the measured part and nominal geometry can make a memorized trajectory unsuitable. In an **illustrative laboratory task**, a robot reads culture plates while preserving identity, sterility, and timing. A reading linked to the wrong plate is a task failure even if the sensor value looks plausible. In the **illustrative home task**, Eva's assistant must interpret an underspecified request and adapt to changing objects, routes, and human behavior. The common problem is selecting and controlling motions that satisfy the *current* task under the *current* constraints, then producing evidence of the result.

2. What current AI shows, and what remains to be established · slides 11–19

Demonstrations and learned proposals

The lecture shows robots unloading laundry, performing coordinated sword movements, and driving. These are demonstrations of particular systems in particular settings. To reason about transfer, change one condition: use another garment, shift a dancer's position, or change a road condition. A success video provides evidence that the visible episode happened. Even many successful videos do not, by themselves, specify the range of tasks, bodies, and worlds in which the system will succeed.

A generative model learns patterns in examples and produces a plausible continuation. The continuation can be text, an image, a video frame, or an action. Plausibility is valuable for producing candidates, but a candidate physical action can affect the world before a person can inspect it. We therefore distinguish **proposal generation** from **commitment checking**. A model may propose carrying Eva's white mug. Before committing to that carry, the robot must establish that the mug is accepted, the route and grasp are feasible, and the relevant safety conditions have enough support for this action.

In one common vision–language–action arrangement, a policy takes camera images, an instruction, and robot state, then produces a short *action chunk*. The robot executes it, observes again, and proposes another chunk. Human teleoperation can supply demonstrations for training. This can yield impressive skills and useful generalization. It does not make a policy “unreasoning,” and it does not imply that every policy lacks an internal prediction of consequences. The scientific question is more specific: **which changes can the trained system handle, which failures occur, and what evidence warrants its next action in this case?**

The changed mug case

Suppose Eva has accepted the white mug. The robot previously served a half-full mug on a particular route without spilling. Now the same mug is nearly full. Keep the robot, route, and turn unchanged. The old success does not decide what will happen next, because the free space above the liquid has changed. A useful account predicts a possible spill *before* the new run and identifies the assumption that the earlier case left untested.

This is the lecture's small lesson in scope. A program does not simply “work” or “fail” in the abstract. Its performance relates to a **problem**, a **body**, and a **world**. Change the mug fill, gripper, route, or recipient, and the relevant claim may change. The classroom D2 program uses invented toy scores to make the comparison visible; it is **TOY MODEL — illustrative**, not a physical spill simulator or measured robot result.

Prediction and imitation can cooperate. A learned policy may propose a carry, while a learned or explicit model predicts whether that carry is likely to meet the task constraints. An explicit model can be wrong; a learned policy may contain useful predictive structure. What matters is whether the system's expectations and limits can be stated, tested, and revised when observations disagree.

From models to testable methods

For the coffee case, a *model* describes relevant relations: how the liquid may move, whether the gripper can hold the mug, and how a handover affects Eva. A *representation* records the current accepted mug, estimated fill, route, Eva's location, and uncertainty. These make *constraints* visible to an *algorithm*: reach, grip, spill margin, route clearance, and safe handover. The planner and controller then choose and regulate an action. The chain is useful only if its predictions are checked against physical episodes.

Building a working program is one constructive test of an explanation. A successful run alone cannot show that the proposed mechanism caused the success; the program might have succeeded by luck or for a different reason. Conversely, a sound component can fail because another part of the system is wrong. Changed cases are valuable because they force a model to predict *where* a method should still work and *where* it should fail. Executable programs also let other people inspect, run, and vary a method.

The diagram on slide 18 connects cognitive and brain science with robotics and AI engineering through a **machine-executable computational model of embodied agency**. One side asks how people perceive, predict, and act; the other asks how to build systems that accomplish tasks. Both can test a model against observations and interventions. The references to internal models, intuitive physics, and semantic digital twins show a line of related ideas; they do not claim that these theories are identical. In Eva's case, a computational model should help predict what a fuller mug or a changed handover will do and support a program that can act accordingly.

The lecture uses five questions to make “understanding” operational. They are a guide, not an exhaustive definition:

Question	Applied to Eva's coffee	What it tests
Explain	Was the mug dry because of a slow turn or because it was half-full?	A causal account of the observed result.
Predict	Will a nearly full mug spill on the same route?	A claim made before the changed episode.
Design	Would another gripper or a tray improve the carry?	Whether the account guides intervention.
Bound	Which mugs, fill levels, routes, and robot bodies are covered?	The scope and limits of a competence claim.
Check	Is there evidence of Eva's acceptance, the chosen action, and the outcome?	Whether another person can assess the episode.

The same questions transfer to stator insertion or laboratory plate handling. For example, a force trace may help *check* an insertion, but the trace alone does not *explain* why the part seated correctly. A useful explanation also identifies a mechanism and makes predictions about changed parts or contact conditions.

3. One physical choice reveals the system problem · slides 20–23

The egg grasp

To grasp an egg, the robot must apply enough force to prevent slipping and little enough to avoid cracking it. Shell strength, contact location, gripper material, and motion can change the safe range. The colored regions on slide 20 are illustrative, not measured force limits. A force that worked once may fail on the next egg. The robot needs observations during contact, a way to adjust the grip, and a condition under which it stops or asks for help. This small case has the same structure as Eva's mug: the current object and current world matter, and sensing must stay connected to action.

What an integrated system has to do

Eva's request lets us assemble the parts of an **integrated intelligent system**. It has to interpret the request, ground words in the current scene, maintain a representation of relevant state, generate possible actions, predict effects, check whether a proposal is supported, command the body, monitor the physical process, respond to deviations, and record what happened. These functions interact. A perfect controller cannot make delivery correct if the selected mug is wrong. A correct plan cannot prevent a collision if the robot fails to monitor a person who moves into its path.

The classroom D1 and D3 programs illustrate two of these distinctions. D1 turns an underspecified request into a decision to ask about the mug. D3 checks whether an action proposal has declared support. Both are **TOY MODEL — illustrative** teaching programs. Their output is a worked logical example, not a robot safety certificate.

Consider the proposal “carry the white mug to Eva.” The lecture's **commitment check** asks six kinds of question:

1. **Grounding:** Is this the drink and mug Eva accepts? If acceptance is unknown, ask.
2. **Feasibility:** Can this robot grasp and carry the current mug along a usable route? A tray changes the effective body and needs its own feasibility check.
3. **Safety:** Are the relevant spill, collision, temperature, and human proximity conditions adequately addressed for this action?
4. **Monitoring:** What observations will tell the system that the carry or handover is departing from expectations?
5. **Repair:** If Eva stands or the route becomes blocked, what will the robot do before continuing?
6. **Recording:** What request, assumptions, choice, observations, and outcome will be available for later assessment?

The check compares **available evidence** with **declared criteria** for this action. Passing it does not mean uncertainty has disappeared. It means the evidence meets the stated threshold for making this commitment. Asking, observing, carrying, pausing, and handing over have different consequences, so the support required for each can differ. If a critical fact remains unresolved, asking or abstaining may serve Eva's goal better than moving immediately.

A record of the episode makes later checking possible; it is not itself an explanation. A critic still needs to compare the record with the task criterion, inspect failures and repairs, and decide what the evidence supports. This is how an episode can contribute to a larger claim about competence without being mistaken for that claim.

For Eva's accepted white mug, keep six stages separate:

Stage	Question or product in this case
Proposal generation	A policy or planner suggests carrying the white mug.
Commitment checking	Does the available evidence meet the declared criteria for *this* mug, route, and handover?
Physical episode	The robot acts, observes the carry, and responds if Eva or the route changes.
Episode record	The request, assumptions, chosen action, observations, and outcome are available for inspection.
Episode verdict	The observed outcome is judged against the declared success criterion for this run.
Competence claim	Results from varied, checked episodes support a stated range of mugs, routes, people, bodies, and world conditions.

A passed commitment check means the available evidence meets the declared criterion for this action; uncertainty remains. A successful episode verdict concerns one run. Neither statement alone establishes the wider competence claim.

---

4. Models and methods across the course · slides 24–29

The integration diagram is the map of the semester. Lecture 2 unpacks the body motion problem. Lecture 3 studies what makes an episode and a competence claim judgeable. Lecture 4 examines the body–world system and its physical constraints. Lecture 5 develops predictive digital twins. Lecture 6 studies how an executive makes, monitors, and revises commitments. Later lectures develop proposal generation, plans, action descriptions, perception, motion control, knowledge, and learning. Each topic supplies part of the path from request to justified world change.

A **digital twin** is a model used to predict possible consequences under stated assumptions. Before opening Eva's dishwasher, the robot might use one to predict the door's motion, the reach to its handle, and whether the open door will obstruct her walking frame. Simulation does not guarantee what the physical world will do. A useful twin produces predictions that observations can test; disagreement should update the model or interrupt the action.

A video of a person manipulating an object can suggest an action plan. To use it, a robot must infer more than a visible path: the object's role, action phases, grasp, intended effect, and conditions that the camera may not reveal. Such a plan is a **candidate**. It must be grounded for this robot, object, and room. A **generalized plan** can retain task structure such as “find a reachable view, grasp appropriately, place at a suitable destination” while resolving exact poses, grasps, and destinations in each new setting. Reuse therefore requires fresh perception and body-specific feasibility checks.

The long-term research goal is a robot that can explore, observe examples, read instructions, form testable accounts of everyday actions, and record predictions and outcomes. The “robot scientist” on slide 29 is a research vision, not a finished system. Its value as a goal is that it connects doing the task with explaining how and when the method works.

The proposed Virtual Research Building work would let students compare predictions and outcomes across changed cases. **TODO(MB):** verify the entry link, account method, first task, and available recording features before assigning that lab. The offline setup check is ready: from the Lecture 1 folder, run `python3 tools/check_env.py` (on Windows, `py tools/check_env.py`). It reports whether the local Python starter is ready and gives a support code if help is needed. A warning about an unchecked browser URL or optional notebook does not mean Python failed. A working Python installation is a practical starting point for the exercises; it is not evidence of robot competence.

---

5. What a success can support · slides 30–32

Return to the opening questions. What can robots do today? We can describe demonstrated capabilities and the conditions in which they were shown. What does a generative model return? It can return a valuable candidate continuation, including an action proposal, whose suitability for this case still has to be checked. What would one hundred successful videos show? They can provide strong evidence for the conditions represented in those runs. They do not automatically establish reliable performance across different tasks, bodies, homes, people, or failures.

A **competence claim** states the task, body, and world families and their ranges; the property to be achieved; how broadly it is claimed; and the evidence status. A success criterion judges each episode. A wider claim needs tests that vary relevant conditions and report failures as well as successes. A checked record can support a verdict on one episode; varied verdicts may support a bounded competence claim. The claim cannot be stronger than its evidence.

Eva's goal remains the final test. If the mug is fuller, the robot might use a slower route or a tray. If Eva stands, it might pause and re-ground the handover. Either revision is only a candidate until its likely effects, constraints, monitoring, and evidence have been considered. The course will ask repeatedly how a robot can pursue a person's goal **flexibly and reliably as the situation changes**, and how we can tell when it has done so.

Rehearse the lecture

Try these without looking back, then use the short answer guide below. For a fuller rehearsal, explain each answer aloud with a changed case.

1. Eva's blue mug is unavailable. Why is using the white mug a grounding question rather than a mere object detection problem?
2. A teleoperator sets a table with a robot. What does this show, and what would still have to be established for autonomy?
3. A learned policy proposes the same turn for a half-full and a nearly full mug. State one prediction and one observation that could test transfer.
4. Apply **explain, predict, design, bound, check** to a changed stator segment or culture plate, rather than to coffee.

5. Give one example in which using a thing changes the robot's effective embodiment. Name the new constraint it introduces.
6. For Eva's accepted white mug, distinguish a *proposal*, a *commitment check*, a *physical episode*, an *episode verdict*, and a *competence claim*.
7. Why can a digital twin help even though its simulation may be wrong?
8. A video shows one hundred successful deliveries. State a defensible narrow claim and the additional evidence needed for a wider one.

Short answer guide

1. Detection establishes that a white mug exists. Grounding establishes whether it satisfies Eva's intended request and constraints; her acceptance may be needed.
2. It supports physical feasibility with the operator's information and judgment in that setting. Autonomous sensing, task interpretation, control, and performance across declared variations remain to be tested.
3. Predict that the nearly full mug may spill during the old turn; observe fill, motion, and any spill on a carefully checked changed episode. The lecture's toy score is not physical evidence.
4. A good answer gives five *different* questions: cause, changed-case outcome, intervention, covered range, and inspectable evidence.
5. A tray may allow a different carry. It introduces tray grasp, stability, clearance, and spill constraints that must be checked.
6. The proposed carry is a candidate; the check assesses it against criteria; the episode is the real action and observations; the verdict judges that episode; the competence claim concerns a declared family of such problems under evidence-supported conditions.
7. It makes consequences explicit before action. Sensor observations can test its prediction and trigger revision or stopping.
8. The runs support success under their demonstrated conditions. A wider claim needs a declared range, varied tests, checked outcomes, and failure evidence.

Terms to keep distinct

Goal — the wanted outcome. **Grounding** — resolving a request in the current situation. **Proposal generation** — producing a candidate action. **Commitment checking** — comparing its evidence with declared criteria. **Monitoring** — checking expectations during execution. **Repair** — revising action after a deviation. **Episode record** — inspectable decisions, actions, observations, and outcomes. **Episode verdict** — judging one run against criteria. **Competence claim** — an evidence-supported statement about a declared family and range of cases.

For the next lecture: Read the AICOR book's preface, framework overview, and opening of Chapter 1; work through Exercise Sheet 1. The VRB task requires verified course access instructions. In your answers, keep separate the requested outcome, the motion chosen, the physical result, and the evidence for claiming success.